

Nearest neighbour interpolation



1

a

i 8 mm

ii 17 mm

iii 7 mm

iv 13.5 mm

b At $(0,0)$ since we would expect it to rain 17 mm there, whereas as at $(3.5, -4)$ we would expect it to rain 12.5 mm since it is on the edge between the regions for C and D .

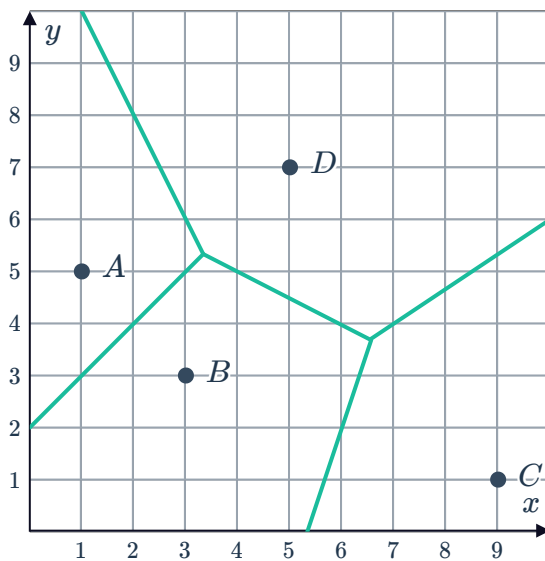
2

a 18°C

b 18°C

3

a



b

i Yes

ii No

iii Yes

iv No

c

i No

ii Yes

iii Yes

iv Yes

4 Point chosen should be in region E and within 5 units of the point $(-3, -4)$.

Example integer answers: $(-1, -1), (-1, 0), (0, 0), (0, -1), (0, -2), (1, -1), (1, -2), (1, -3)$

5

a 0.25 inches

b 0.27 inches

c 0.26 inches

d 0.43 inches



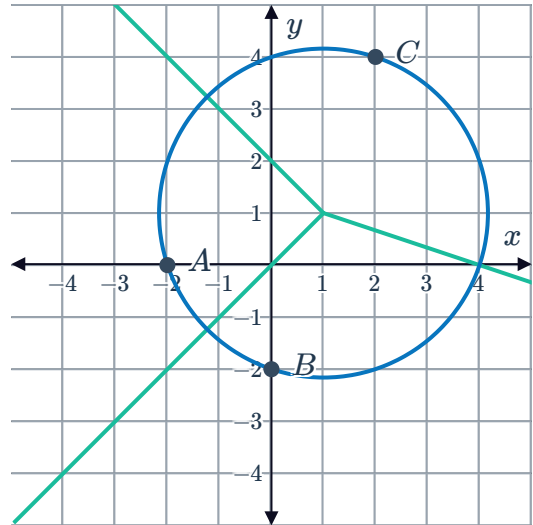
Largest empty circle

6

a

i Centre: $(1, 1)$, radius: 3.16

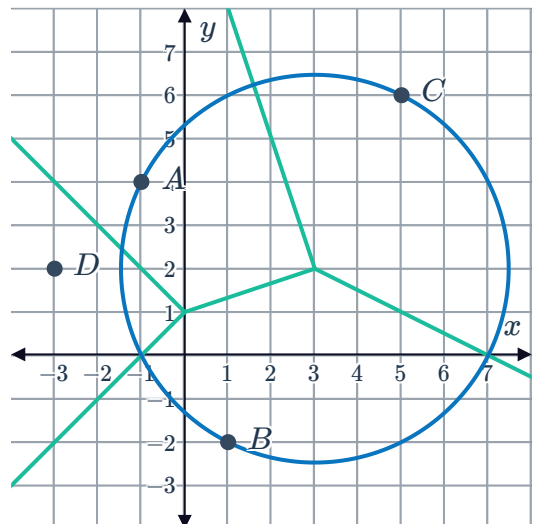
ii



b

i Centre: $(3, 2)$, radius: 4.47

ii



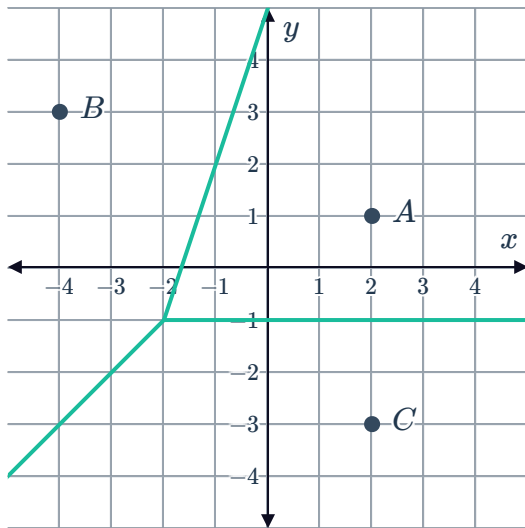
7 Centre: $(-3, 5)$ or $(5, -3)$, radius: $2\sqrt{10}$

8



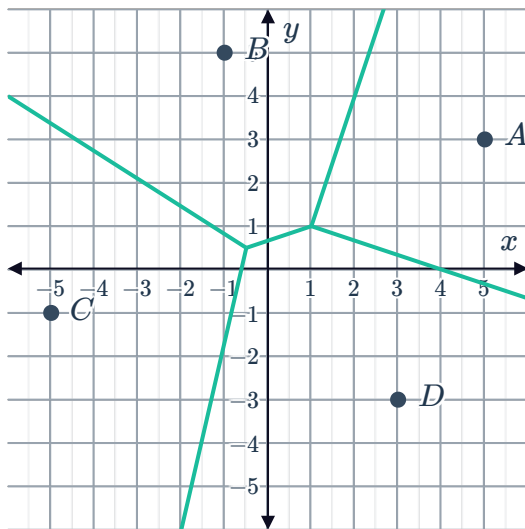
a

i

ii Centre: $(-2, -1)$, radius: 4.472

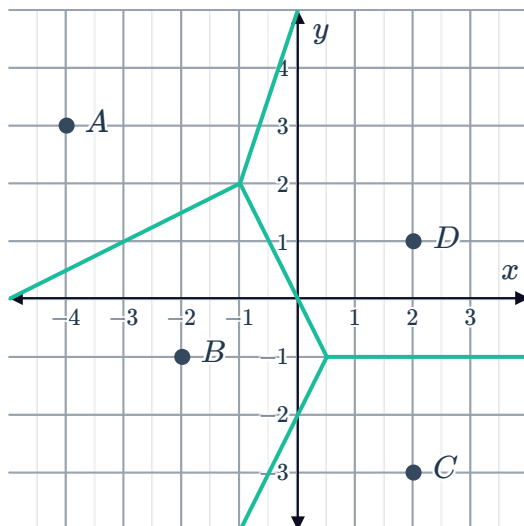
b

i

ii Centre: $(-0.5, 0.5)$, radius: 4.528

9

a

b $(-1, 2)$